Surface-Mount TMBS® (Trench MOS Barrier Schottky) Rectifier



SMB (DO-214AA)

Cathode ○—◀—○ Anode

FEATURES

- Low profile package
- Ideal for automated placement
- Trench MOS Schottky technology
- Low power losses, high efficiency
- Low forward voltage drop
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT

TYPICAL APPLICATIONS

For use in low voltage, high frequency inverters, freewheeling, DC/DC converters, and polarity protection applications.

MECHANICAL DATA

Case: SMB (DO-214AA)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS-compliant, commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
E3 suffix meets JESD 201 class 2 whisker test

Polarity: color band denotes the cathode end

LINKS TO ADDITIONAL RESOURCES

3D Models

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	4.0 A
V_{RRM}	100 V
I_{FSM}	80 A
E_{AS}	50 mJ
V_F at I_F = 4.0 A	0.61 V
T_J max.	150 °C
Package	SMB (DO-214AA)
Circuit configurations	Single

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER	SYMBOL	VSSB410S	UNIT
Device marking code		V4B	
Maximum repetitive peak reverse voltage	V_{RRM}	100	V
Maximum DC forward current	I_F (1)	4.0	A
	I_F (2)	1.9	
Peak forward surge current 10 ms single half sine-wave superimposed on rated load	I_{FSM}	80	A
Non-repetitive avalanche energy at T_J = 25 °C, L = 60 mH	E_{AS}	50	mJ
Peak repetitive reverse current at t_p = 2 µs, 1 kHz, T_J = 38 °C ± 2 °C	I_{RRM}	1.0	A
Operating junction and storage temperature range	T_J, T_{STG}	-40 to +150	°C

Notes

(1) Mounted on 14 mm x 14 mm pad areas, 1 oz. FR4 PCB
(2) Free air, mounted on recommended copper pad area

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Breakdown voltage	I_R = 1.0 mA	T_A = 25 °C	V_{BR}	100 (minimum)	-	V
Instantaneous forward voltage	I_F = 4.0 A	T_A = 25 °C	V_F (1)	0.68	0.77	V
		T_A = 125 °C		0.61	0.69	
Reverse current	V_R = 70 V	T_A = 25 °C	I_R (2)	1.5	-	µA
		T_A = 125 °C		1.2	-	mA
	V_R = 100 V	T_A = 25 °C		7.0	250	µA
		T_A = 125 °C		3.6	20	mA
Typical junction capacitance	4.0 V, 1 MHz		C_J	230	-	pF

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 40 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)			
PARAMETER	**SYMBOL**	**VSSB410S**	**UNIT**
Typical thermal resistance	$R_{\theta JA}$ (1)	120	°C/W
	$R_{\theta JM}$ (2)	15	

Notes

(1) Free air, mounted on recommended PCB 1 oz. pad area. Thermal resistance $R_{\theta JA}$ - junction to ambient

(2) Units mounted on PCB with 14 mm x 14 mm copper pad areas. $R_{\theta JM}$ - junction to mount

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PREFERRED PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
VSSB410S-E3/52T	0.096	52T	750	7" diameter plastic tape and reel
VSSB410S-E3/5BT	0.096	5BT	3200	13" diameter plastic tape and reel

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

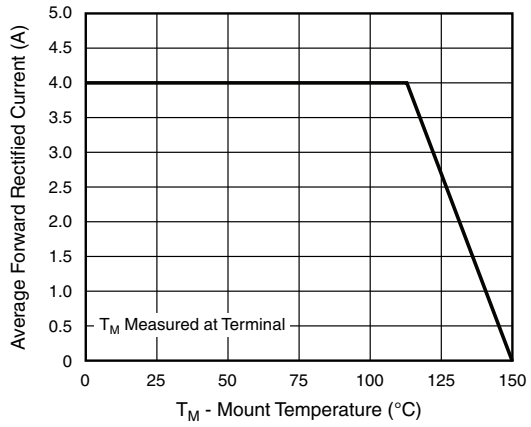


Fig. 1 - Maximum Forward Current Derating Curve

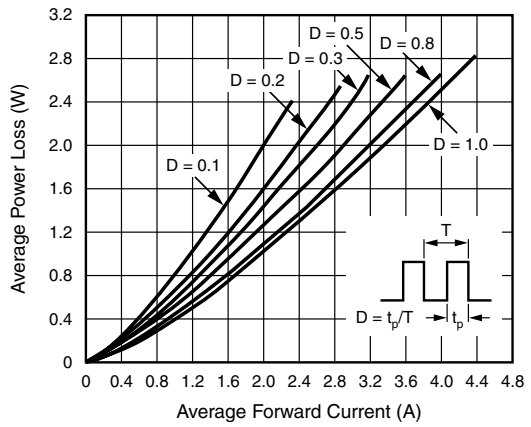


Fig. 2 - Forward Power Loss Characteristics

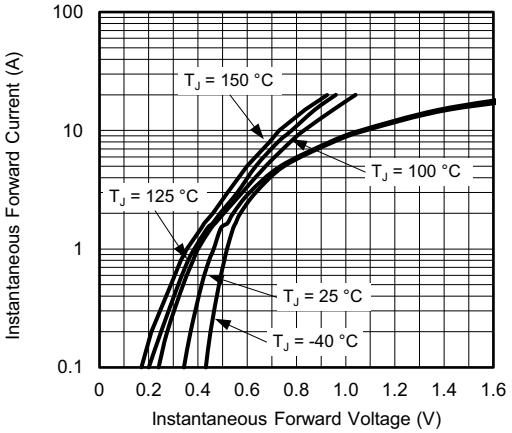


Fig. 3 - Typical Instantaneous Forward Characteristics

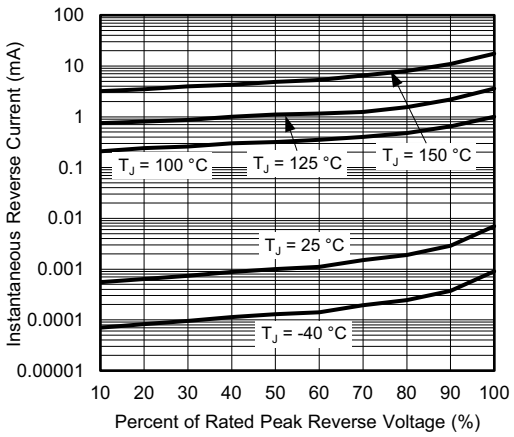


Fig. 4 - Typical Reverse Characteristics

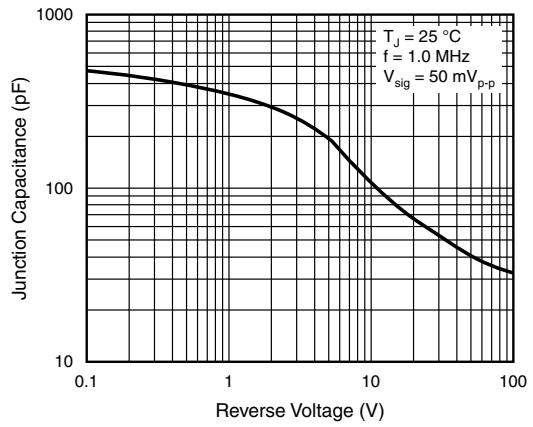


Fig. 5 - Typical Junction Capacitance

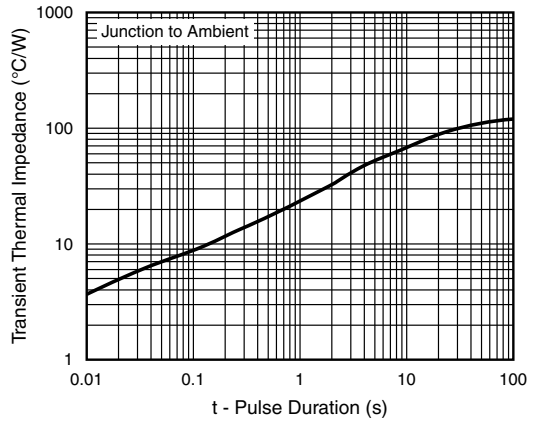
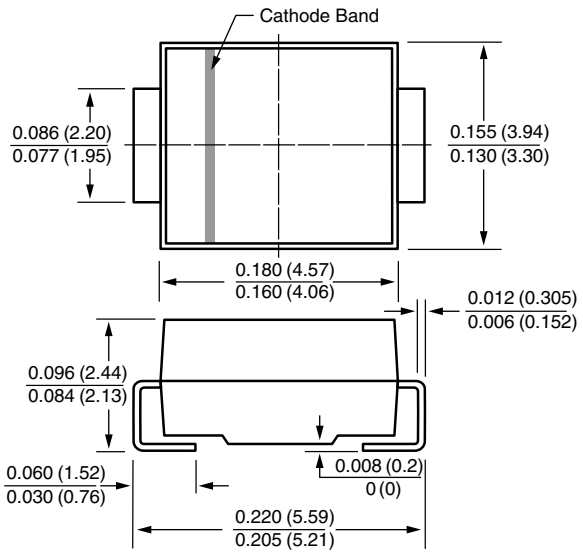


Fig. 6 - Typical Transient Thermal Impedance

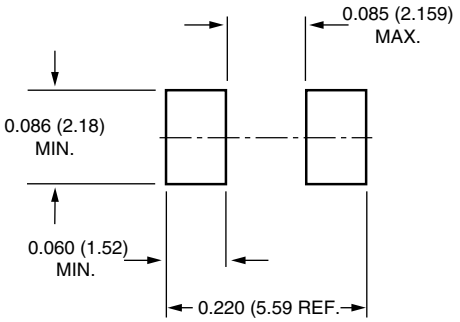
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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

SMB (DO-214AA)



Mounting Pad Layout



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